

THE EXPECTED VALUES OF THE ELEMENTARY SYMMETRIC FUNCTIONS OF SOME MATRICES

D.J. de Waal*

University of the Orange Free State

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SUMMARY

This paper is concerned with the expected values of the elementary symmetric functions (esf's) of matrices that are often used in multivariate analysis. In de Waal (1972) and Saw (1973) the expected values of the esf's of the central Wishart (general Σ) and the noncentral Wishart ($\Sigma = \sigma^2 I$) are given in an explicit form. Krishnaiah and Chattopadhyay (1975) however gave an expression for the s -th moment of the esf's for a wide class of matrices, but the result is limited since the expression is given as an infinite series and special cases are difficult to consider. The results in this paper are in terms of finite series and easy to use.

The specific matrices that will be considered are (i) noncentral Wishart (general Σ), (ii) matrices in the MANOVA case, (iii) matrices in the GMANOVA case and (iv) the matrix in the canonical correlation case.

1. Wishart case

From de Waal (1972) and Saw (1973) we have that if $B(p \times p)$ is distributed $W(\Sigma, q, \Omega)$

$$(1.1) \quad E \operatorname{tr}_j B = \begin{cases} (q)^{(j)} \operatorname{tr}_j \Sigma & \text{if } \Omega = 0 \\ \sigma^{2j} \sum_{i=0}^j (q-1)^{(j-1)} (p-i)^{(j-1)} \operatorname{tr}_i \Omega & \text{if } \Sigma = \sigma^2 I_p \\ |\Sigma| \sum_{i=0}^p (q-i)^{(p-i)} \operatorname{tr}_i \Omega & \text{if } j = p \end{cases}$$

where $\operatorname{tr}_j B$ denotes the j -th esf of B ; Ω the noncentrality parameter and $(q)^{(j)} = q(q-1) \dots (q-j+1)$, $\operatorname{tr}_0 B = 1$.

* Part of this research has been done while visiting the University of North Carolina at Chapel Hill and Stanford University.

For general Σ the result can be obtained as follows:

Let $X(p \times q)$ be distributed $N(M, \Sigma \otimes I_q)$ where M is $p \times q$, then for $q \geq p$ $B = XX'$ is distributed $W(\Sigma, q, \Omega)$ with $\Omega = \Sigma^{-1}MM'$. If $X'_J = (X'_J X'_J)$ then $X_J(j \times p)$ is distributed $N(M_J, \Sigma_{JJ} \otimes I_q)$ where $M' = (M'_J, M'_{\bar{J}})$ and

$$\Sigma = \begin{bmatrix} \Sigma_{JJ} & \Sigma_{J\bar{J}} \\ \Sigma_{\bar{J}J} & \Sigma_{\bar{J}\bar{J}} \end{bmatrix} \begin{matrix} j \\ p-j \end{matrix}$$

Therefore $B_{JJ} = X_J X'_J$ is distributed $W(\Sigma_{JJ}, q, \Sigma_{JJ}^{-1} M_J M'_J)$

Since $\text{tr}_j B = \Sigma_j |B_{JJ}|$ where $\Sigma_j |B_{JJ}|$ denotes the summation over all principal minors of order j of B , it follows from (1.1) that:

$$\begin{aligned} (1.2) \quad E \text{tr}_j B &= \Sigma_j E |B_{JJ}| \\ &= \Sigma_{i=0}^j (q-i) \binom{j-i}{i} \gamma_i^j(\Sigma, \Omega) \end{aligned}$$

$$\text{where} \quad \gamma_i^j(\Sigma, \Omega) = \Sigma_j |\Sigma_{JJ}| \text{tr}_i \Sigma_{JJ}^{-1} M_J M'_J$$

This result has also been derived by Shah and Khatri (1974).

2. Manova case

Suppose $A(p \times p)$ is distributed independently of B according to $W(\Sigma, n)$, then $V = A^{-1}B$ or written as $V = B^{1/2} A^{-1} B^{1/2}$ if we talk of the distribution of V , is a well-known matrix in the MANOVA case. B is known as the variance due to hypothesis and A due to error. If $q < p$, then B is singular and the equivalent of V is then defined as

$$(2.1) \quad F = X' A^{-1} X$$

where X is defined in section 1. We shall first consider $E \text{tr}_j F$. The density of $F(q \times q)$ is derived by James (1964) and is given by

$$(2.2) \quad \frac{\Gamma_q(\frac{1}{2}(n+q))}{\Gamma_q(\frac{1}{2}(n+q-p)) \Gamma_q(\frac{1}{2}p)} \text{etr}(-\frac{1}{2}\Omega) |F|^{\frac{1}{2}(p-q-1)} |I+F|^{-\frac{1}{2}(n+q)} {}_1F_1(\frac{1}{2}(n+q);$$

$$\frac{1}{2}p; \frac{1}{2}\Omega F(I+F)^{-1}), F > 0$$

It is called the noncentral multivariate beta density of the second kind with p and $n+q-p$ degrees of freedom and noncentrality parameter $\Omega = M' \Sigma^{-1} M$. The ${}_1F_1$ function is called the confluent hypergeometric function of matrix argument and

$$(2.3) \Gamma_q(a) = \pi^{q/2} q^{(q-1)} \prod_{i=1}^q \Gamma(a + \frac{1}{2}(1-i))$$

The following lemmas are useful:

Lemma 2.1

Let $\Omega(q \times q)$ be symmetrical positive definite, then (de Waal (1972), Saw (1973), Shah and Khatri (1974))

$$(2.4) \frac{2^q \Gamma_q(\frac{1}{2}a+1)}{\Gamma_q(\frac{1}{2}a)} \text{etr}(-\frac{1}{2}\Omega) {}_1F_1(\frac{1}{2}a+1; \frac{1}{2}a; \frac{1}{2}\Omega) \\ = \sum_{i=0}^q (a-i)^{(q-i)} \text{tr}_i \Omega$$

Lemma 2.2

If $F(q \times q)$ is distributed according to (2.2), then

$$(2.5) E|F| = \frac{1}{(n+q-p-2)^{(q)}} \sum_{i=0}^q (p-i)^{(q-i)} \text{tr}_i \Omega$$

Proof

By direct integration (see de Waal (1969)) it follows that

$$(2.6) E|F| = \frac{\Gamma_q(\frac{1}{2}p+1) \Gamma_q(\frac{1}{2}(n+q-p)-1)}{\Gamma_q(\frac{1}{2}p) \Gamma_q(\frac{1}{2}(n+q-p))} \text{etr}(-\frac{1}{2}\Omega) {}_1F_1(\frac{1}{2}p+1; \frac{1}{2}p; \frac{1}{2}\Omega).$$

By using lemma 2.1 and the identity

$$(2.7) \Gamma_q(\frac{1}{2}a-1)/\Gamma_q(\frac{1}{2}a) = 2^q/(a-2)^{(q)}$$

the lemma follows. (2.7) follows from (2.3).

Theorem 2.1

If $F(q \times q)$ is distributed as a noncentral multivariate beta distribution of the second kind with p and $n-p+q$ degrees of freedom and noncentrality parameter Ω , then

$$(2.8) E \operatorname{tr}_i F = \frac{1}{(n+j-p-2)(j)} \sum_{i=0}^j \binom{q-i}{j-i} (p-i)^{(j-i)} \operatorname{tr}_i \Omega$$

Proof

Let $X = (X_J, X_J^-)$ such that X_J is $p \times j$.

Hence X_J is distributed $N(M_J, \Sigma \otimes I_j)$ where $M_J = (M_J, M_J^-)$.

Since

$$F = \begin{bmatrix} X_J' A^{-1} X_J & X_J' A^{-1} X_J^- \\ X_J'^- A^{-1} X_J & X_J'^- A^{-1} X_J'^- \end{bmatrix} = \begin{bmatrix} F_{JJ} & F_{JJ^-} \\ F_{JJ^-} & F_{JJ'^-} \end{bmatrix}, \text{ say,}$$

then it is clear that F_{JJ} is distributed as a noncentral multivariate beta distribution of the second kind with p and $n+j-p$ degrees of freedom and noncentrality parameter

$$M_J' \Sigma^{-1} M_J.$$

Hence according to lemma 2.2

$$E|F_{JJ}| = \frac{1}{(n+j-p-2)(j)} \sum_{i=0}^j (p-i)^{(j-i)} \operatorname{tr}_i (M_J' \Sigma^{-1} M_J).$$

Since (Saw (1975))

$$\sum_J \operatorname{tr}_i (M_J' \Sigma^{-1} M_J) = \binom{q-i}{j-i} \operatorname{tr}_i \Omega,$$

the theorem follows.

A similar result to (2.8) can be obtained for V . The result follows as a corollary from theorem 2.1 since the density of $B^{\frac{1}{2}} A^{-1} B^{\frac{1}{2}}$ is also a noncentral multivariate beta of the second kind with the following correspondences:

$$q \rightarrow p; n-p \rightarrow q \rightarrow n; p \rightarrow q.$$

Hence

Corollary 2.1 If $V(p \times p)$ is distributed as a noncentral multivariate beta distribution of the second kind with q and n degrees of freedom and noncentrality parameter Ω , then

$$(2.9) E \operatorname{tr}_j V = \frac{1}{(n+j-p-2)(j)} \sum_{i=0}^j \binom{p-i}{j-i} (q-i)^{(j-i)} \operatorname{tr}_i \Omega$$

This result reduces to (1.2) given in Saw (1973) if $n \rightarrow \infty$.

3. Gmanova case

The GMANOVA case can be summarized in the following canonical form (Fujikoshi (1974)):

Let V be defined as in section 2, but with $B(vxv)$ conditional on W be distributed $W(I_v, u, \Delta)$ where $\Delta = D(I+W)^{-1}D$. $D(uxu)$ is a matrix with only the first $s = \min(u, v)$ diagonal elements not equal to zero.

For $p - q \geq u$, $W(uxu)$ is distributed as a central multivariate beta distribution with $p - q$ and $n + u - p + q$ degrees of freedom. $A(vxv)$ is distributed independently of B and W as $W(I_v, n)$.

The following theorem can now be proved:

Theorem 3.1

If $u \geq v$, then unconditionally

$$(3.1) E \operatorname{tr}_j V = \frac{1}{(n+j-v-2)(j)} \sum_{i=0}^j \binom{v-i}{j-i} (u-i)^{(j-i)} (n+q-p+u)^{(i)} \operatorname{tr}_i D^2 / (n+u)^{(i)}$$

Proof:

V conditionally on W is distributed as a noncentral multivariate beta distribution of the second kind with u and n degrees of freedom and noncentrality parameter

$$\Delta = D(I+W)^{-1}D.$$

Therefore, from corollary 2.1

$$(3.2) E_V \operatorname{tr}_j (V|W) = \frac{1}{(n+j-v-2)(j)} \sum_{i=0}^j \binom{v-i}{j-i} (u-i)^{(j-i)} \operatorname{tr}_i \Delta.$$

To obtain the unconditional expectation we note from (2.2) that the density function of W is

$$(3.3) f(w) = \frac{\Gamma_u(\frac{1}{2}(n+u))}{\Gamma_u(\frac{1}{2}(p-q))\Gamma_u(\frac{1}{2}(n-p+q+u))} |w|^{\frac{1}{2}(p-q-u-1)} |I+w|^{-\frac{1}{2}(n+u)}, w > 0.$$

Therefore the expectation of the zonal polynomial $C_K(\Delta)$ for the partition

$$K = (k_1, \dots, k_p)$$

is given by

$$\begin{aligned} (3.4) \quad E C_K(\Delta) &= \int_{w>0} f(w) C_K(D^2(I+w)^{-1}) dw \\ &= \int_{Z>0} f(Z^{-1})|Z|^{-(u+1)} C_K(D^2Z(I+Z)^{-1}) dZ \\ &= \frac{(\frac{1}{2}(n+q-p+u))_K}{(\frac{1}{2}(n+u))_K} C_K(D^2) \end{aligned}$$

The integral w.r.t. Z can be checked from (3.2).

By letting $K = (1^1)$, we have $C_K(D^2) = 2^{k(i+1)-1} \text{tr}_i D^2$ (Shah and Khatri (1974) equation 2.1.2) and the theorem follows.

If $u < v$, u and v are only interchanged in the terms

$$\binom{v-j}{j-1} \text{ and } (u-i)\binom{j-i}{i-1} \text{ in (3.1).}$$

It may be mentioned that the density of $Y = B^{\frac{1}{2}}A^{-1}B^{\frac{1}{2}}$ conditionally on W is

$$(3.5) \quad \frac{\Gamma_v(\frac{1}{2}(n+u))}{\Gamma_v(\frac{1}{2}n)\Gamma_v(\frac{1}{2}u)} \text{etr}(-\frac{1}{2}\Delta)|Y|^{\frac{1}{2}(u-v-1)}|I+Y|^{-\frac{1}{2}(n+u)}$$

$${}_1F_1(\frac{1}{2}(n+u); \frac{1}{2}u; \frac{1}{2}\Delta Y(I+Y)^{-1}), Y > 0.$$

The unconditional density of Y can therefore be obtained by taking the expectation w.r.t. W . This requires the expectation

$$(3.6) \quad E_W \text{etr}(-\frac{1}{2}D^2(I+W)^{-1}) C_K(\frac{1}{2}R(I+W)^{-1})$$

where $R = Y(I+Y)^{-1}D^2$, which seems difficult to obtain in an explicit form. Integrals of this type were considered in Nel and de Waal (1976).

4. Canonical correlation case

The relevant matrix in a canonical correlation analysis is

$$(4.1) \quad V = A^{-1} G$$

where $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ $\begin{matrix} q \\ p-q \end{matrix}$ is distributed $W(\Sigma, n)$

and $A_{11.2} = A_{11} - A_{12}A_{22}^{-1}A_{21}$, $G = A_{12}A_{22}^{-1}A_{21}$. It is known (Troskie (1969)), that $A_{11.2}$ is distributed $W(\Sigma_{11.2}, n-p+q)$ and G conditionally on A_{22} is distributed $W(\Sigma_{11.2}^{-1}, p-q, \Delta^*)$ independently of $A_{11.2}$. $\Delta^* = \Sigma_{11.2}^{-1/2} \beta A_{22} \beta' \Sigma_{11.2}^{-1/2}$ and $\beta = \Sigma_{12} \Sigma_{22}^{-1}$. Σ is partitioned in a similar manner to A .

The following theorem can now be proved:

Theorem 4.1 If $V(q \times q)$ is defined as in (4.1) and

$$P = \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}, \text{ then}$$

$$(4.2) \quad E \operatorname{tr}_j V = \frac{1}{(n+j-p-2)(j)} \sum_{i=0}^j \binom{n}{j-i} \binom{q-i}{j-i} (p-q-i)^{(j-i)} \operatorname{tr}_i P(I-P)^{-1}.$$

Proof:

Since $G^{1/2} A_{11.2}^{-1} G^{1/2}$ conditionally on A_{22} ($p-q \times p-q$) is distributed as a noncentral multivariate beta distribution of the second kind with $p-q$ and $n-p+q$ degrees of freedom with noncentrality parameter $\Delta^* = \Sigma_{11.2}^{-1/2} \beta A_{22} \beta' \Sigma_{11.2}^{-1/2}$, it follows from (2.9) that

$$(4.3) \quad E(\operatorname{tr}_j V | A_{22}) = \frac{1}{(n+j-p-2)(j)} \sum_{i=0}^j \binom{q-i}{j-i} (p-q-i)^{(j-i)} \operatorname{tr}_i \Delta^*$$

But A_{22} is distributed $W(\Sigma_{22}, n)$ and therefore from (1.1)

$$(4.4) \quad E \operatorname{tr}_i (\beta' \Sigma_{11.2}^{-1} \beta A_{22}) = (n)^{(i)} \operatorname{tr}_i (\beta' \Sigma_{11.2}^{-1} \beta \Sigma_{22})$$

Substitute (4.4) for the expectation of $\operatorname{tr}_i \Delta^*$ in (4.3) and the theorem follows.

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